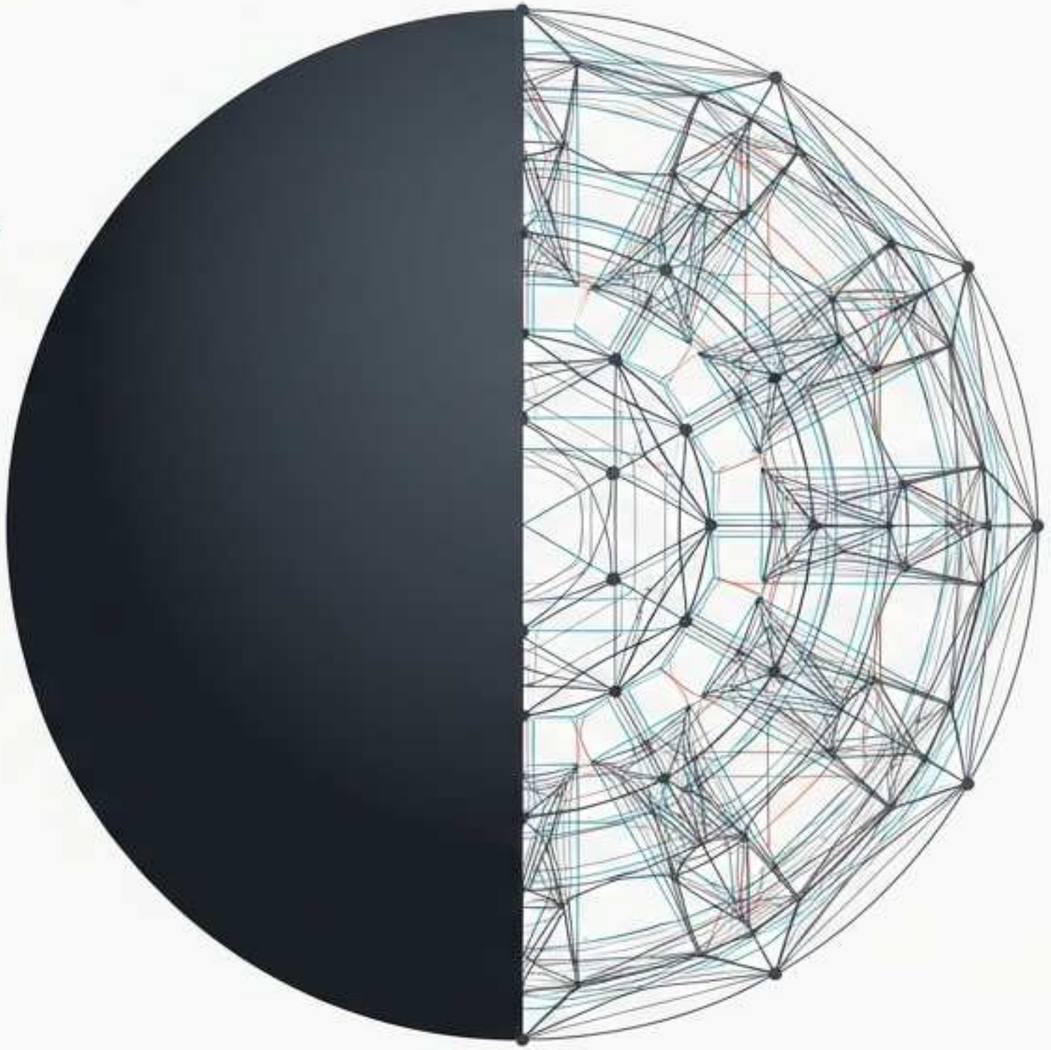
Admissibility and the Structure of Physical Symmetry



From Groups to Groupoids:
Revealing the hidden architecture
of source, transport, and state.

Flyxion, Independent Researcher | 2026

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This framework reinterprets existing mathematics rather than replacing the Standard Model

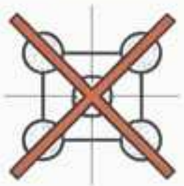
What This Is Not



Not a derivation of the $SU(3) \times SU(2) \times U(1)$ gauge group.



Not a prediction of specific representation content.



Not a competing physical theory to the Standard Model.

What This Is



An interpretive reading of established group and gauge theory.



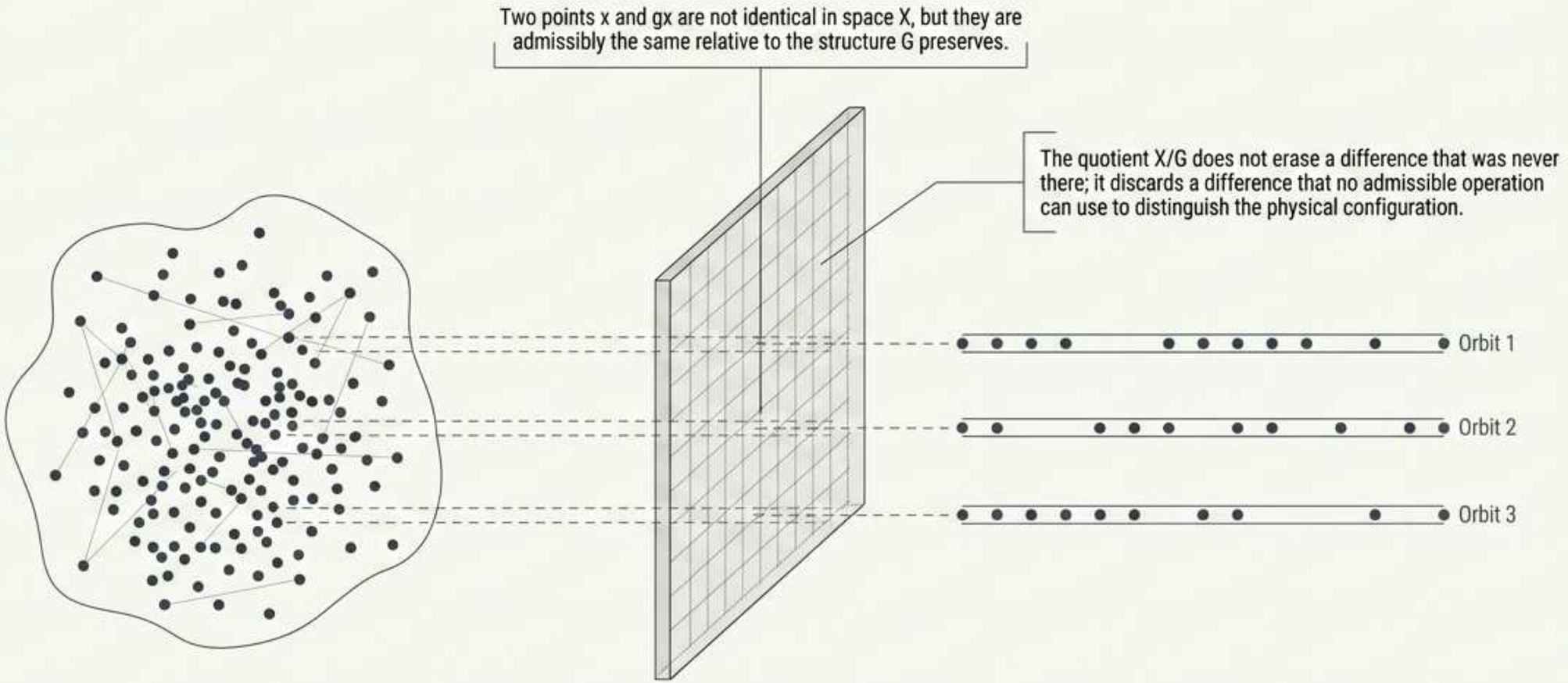
A demonstration that isolated technical facts (quotients, connections, curvature) are instances of a single pattern.



A falsifiable endpoint showing when groupoids can be compressed into groups.

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Symmetry acts as a filter that removes physically inadmissible distinctions



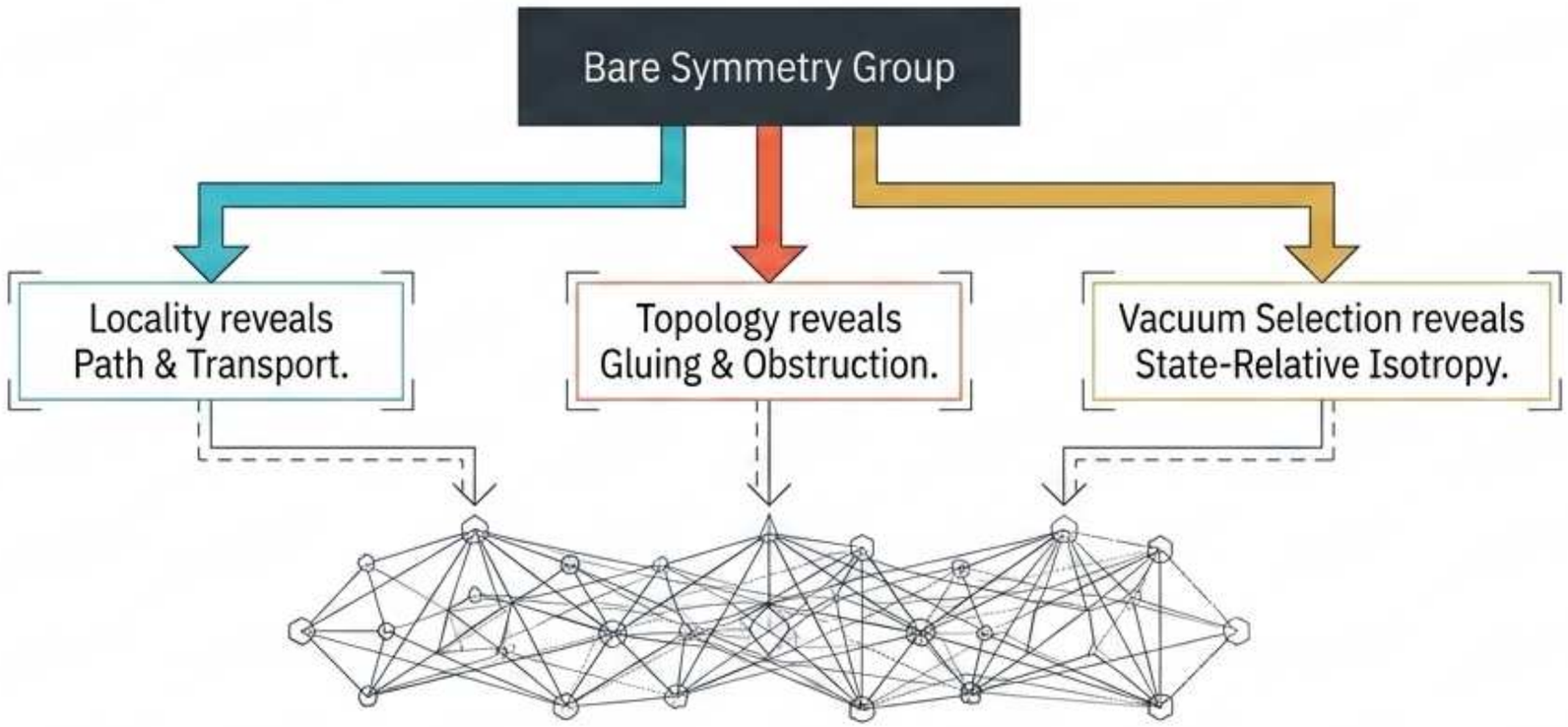
An observable O respects the quotient exactly when $O(gx) = O(x)$ for all g in G.

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A symmetry group is the compression of a richer admissibility groupoid

Physical symmetry admits compression to a group description exactly to the extent that internal data can be suppressed without losing physically relevant distinctions.

Progressive Decompression Arc

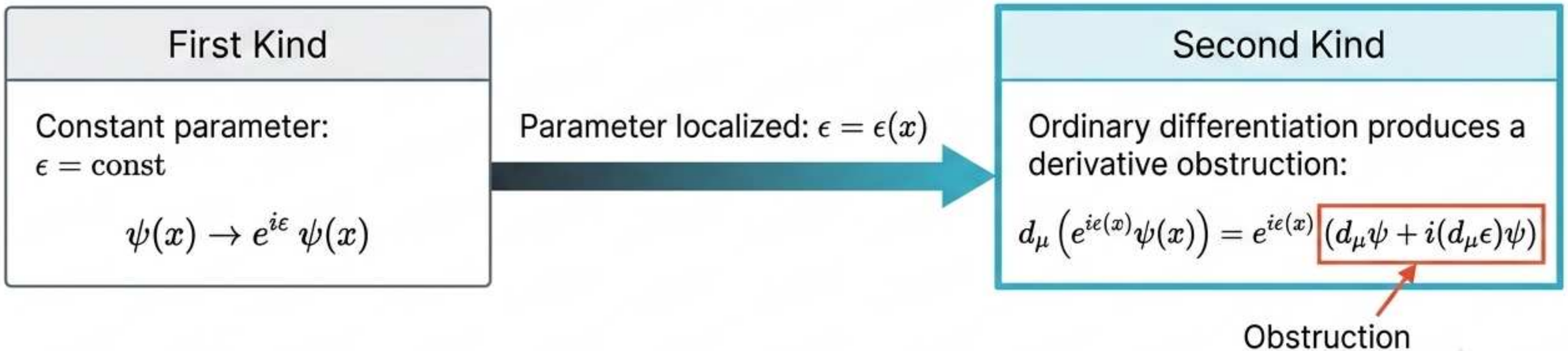


Admissibility Groupoid

When source, target, transport, and state-relative dependence cannot be forgotten, the groupoid retains the necessary data.

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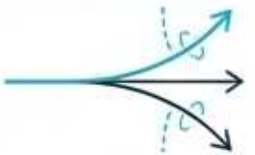
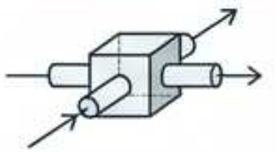
Permitting transformations to vary by position breaks ordinary differentiation



Key Insight: A transformation that is admissible at each point separately does not by itself determine how representatives at neighboring points are compared. Local admissibility does not automatically supply admissible transport.

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The covariant derivative supplies the missing rule of transport

	Global Symmetry	Local Symmetry
Parameter Domain	Constant (ϵ)	Variable by position $(\epsilon(x))$
Derivative Behavior	Preserved (d_μ)	Obstructed by an extra term 
Required Repair Structure	None	Covariant Connection $(D_\mu = d_\mu + ieA_m)$ 
Action on State	Uniform action across the entire domain 	Point-by-point assignment requiring transport rules 

The gauge potential (A_μ) is not an arbitrary complication; it is the exact structure required to relate local realizations once admissibility is indexed by location.

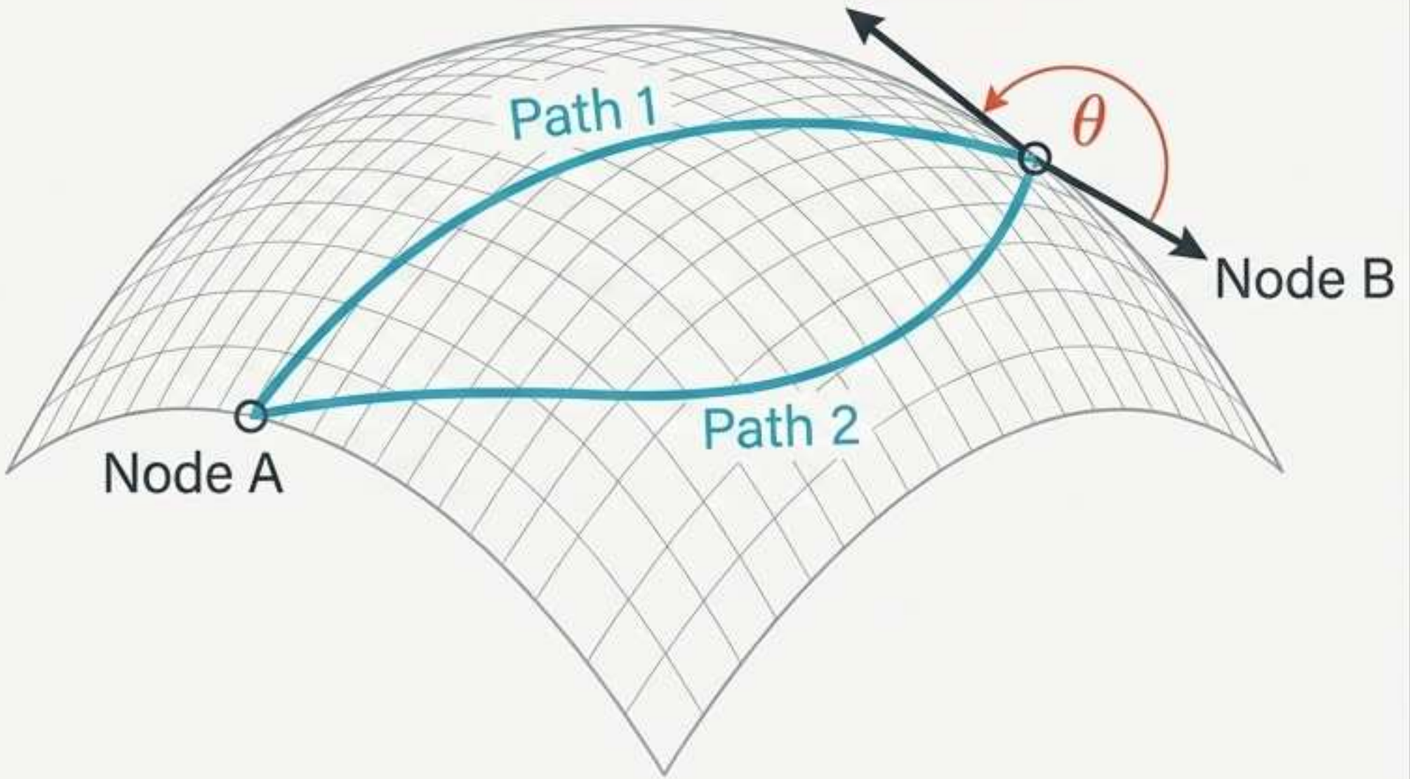
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Curvature proves that local admissibility does not guarantee path-independent global identification

$$F = dA + A \wedge A$$

(Curvature records the failure of transport).

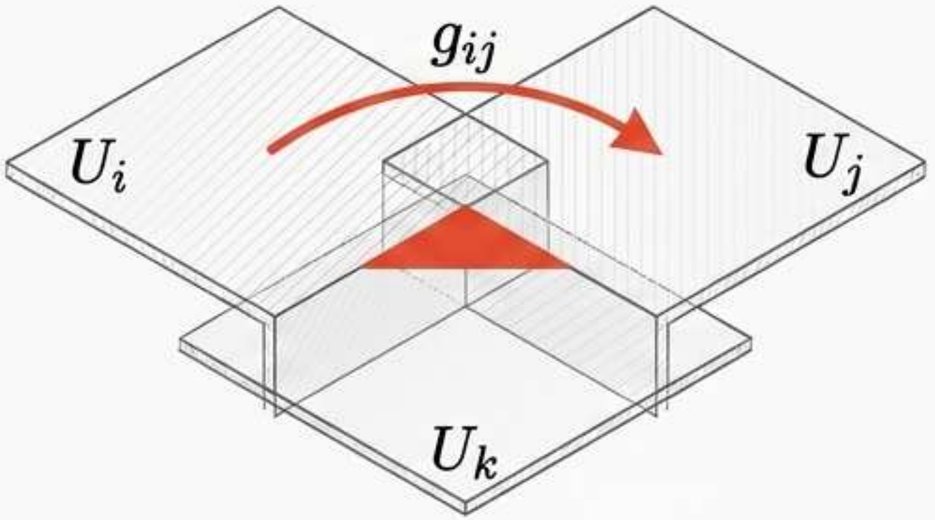
If a representative is transported through locally admissible comparisons, composing those transports depends on the path. Infinitesimal transport around a closed loop need not return trivially.



Trajectory-first Admissibility: Admissibility of each local step does not equal direct equivalence between endpoints. The path history must be retained.

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Locally consistent descriptions require precise transition functions functions to overlap



The Gluing Rule:
On overlap $U_i \cap U_j$, local representatives are related by $g_{ij} : U_i \cap U_j \to G$.

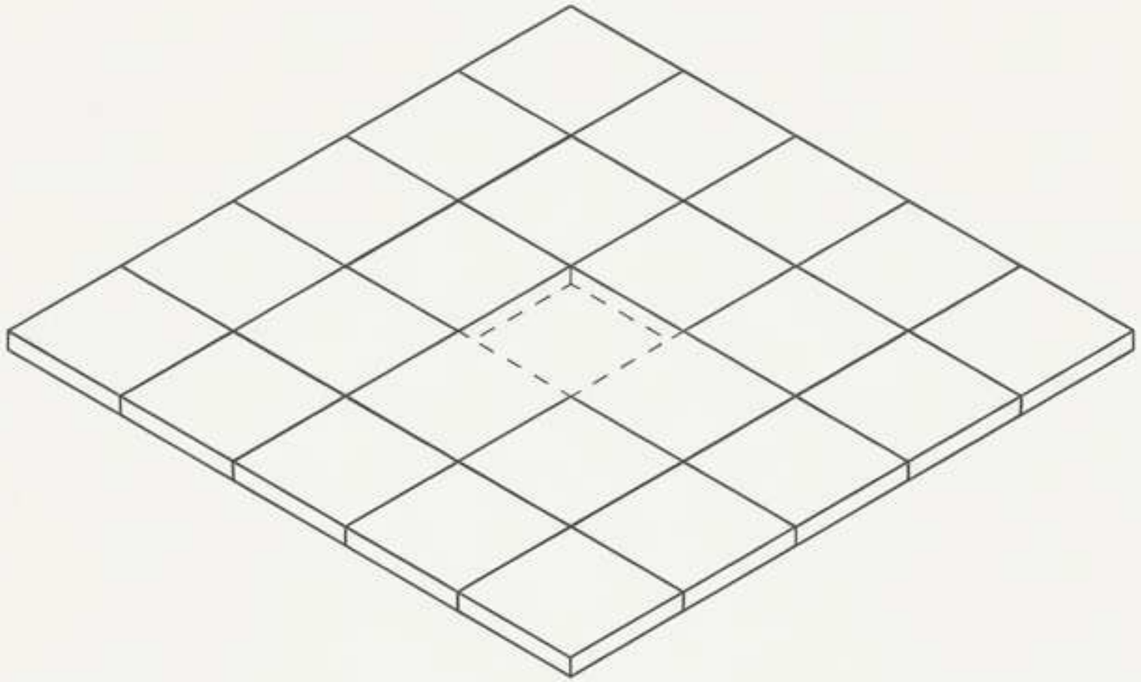
The Cocycle Condition:
$g_{ij} \cdot g_{jk} = g_{ik}$ on $U_i \cap U_j \cap U_k$.

Takeaway:
This is purely local consistency made checkable.
Comparing representatives around any triple overlap gives the same answer regardless of the order of transitions.

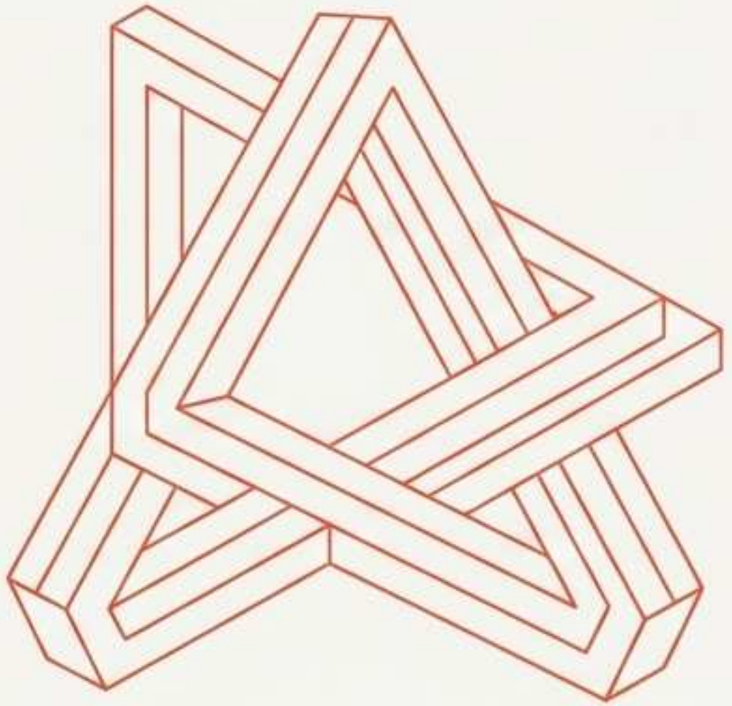
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An obstruction to global gluing is an architectural reality, not an unfinished reconstruction

Unfinished Reconstruction



Nontrivial Obstruction



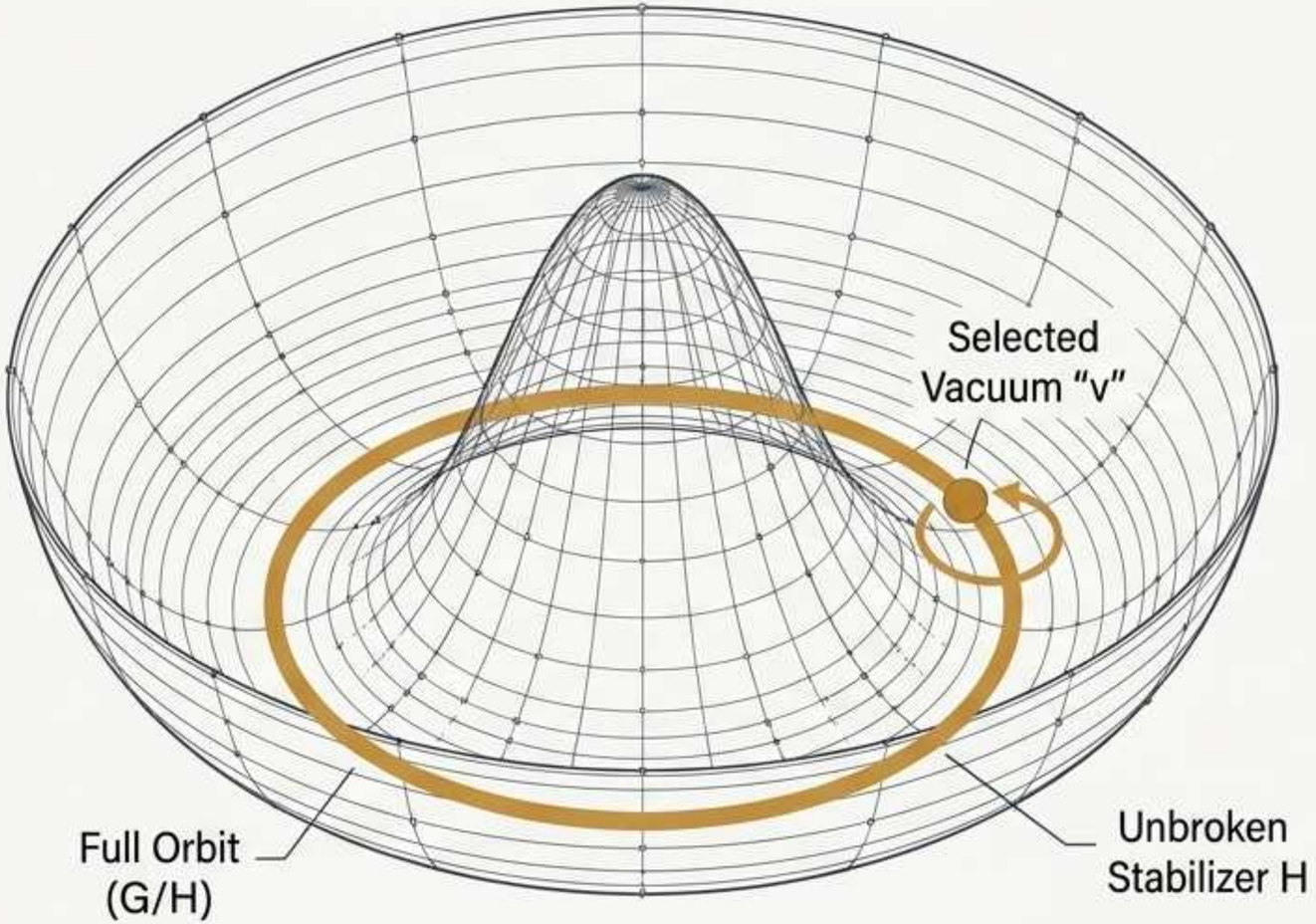
Core Principle: Local consistency does not imply global triviality.

Explanation: A persistent nontrivial obstruction (a nontrivial cocycle class) means no single global representative can stand in for the whole family without discarding local data.

Connection to Admissibility: Not every "stall" is a failure to reconstruct. Sometimes the obstruction is not an unfinished map; it is the authentic, non-global structure of the object itself.

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Symmetry breaking refines constraints without destroying the underlying symmetry



The Landscape of Constraint Refinement

The Stabilizer:

$$H = \mathrm{Stab}_G(v) = \{g \in G : gv = v\}$$

Constraint Refinement:

The dynamics single out a vacuum v. The original G symmetry of the theory remains, but the symmetry of the selected state is refined to H.

The Orbit:

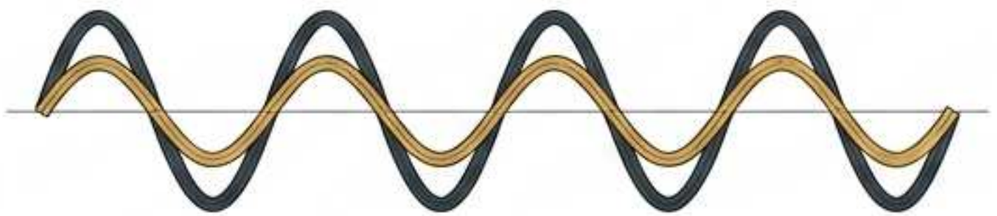
For generators outside H, the state v is carried along the gold ring (G/H) to distinct, degenerate vacua.

Takeaway:
Equivalence must now be evaluated not just by the action of G, but relative to the preservation of the specifically selected vacuum state.

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The physical realization of broken directions **depends on the surrounding compositional architecture**

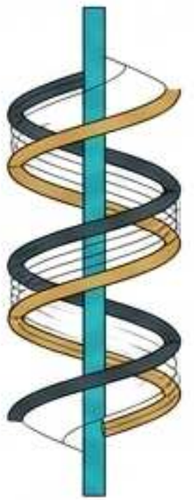
Global Symmetry



Goldstone Theorem

Global Fate: Broken directions survive as independent, physically propagating massless Goldstone modes.

Local Gauge Symmetry



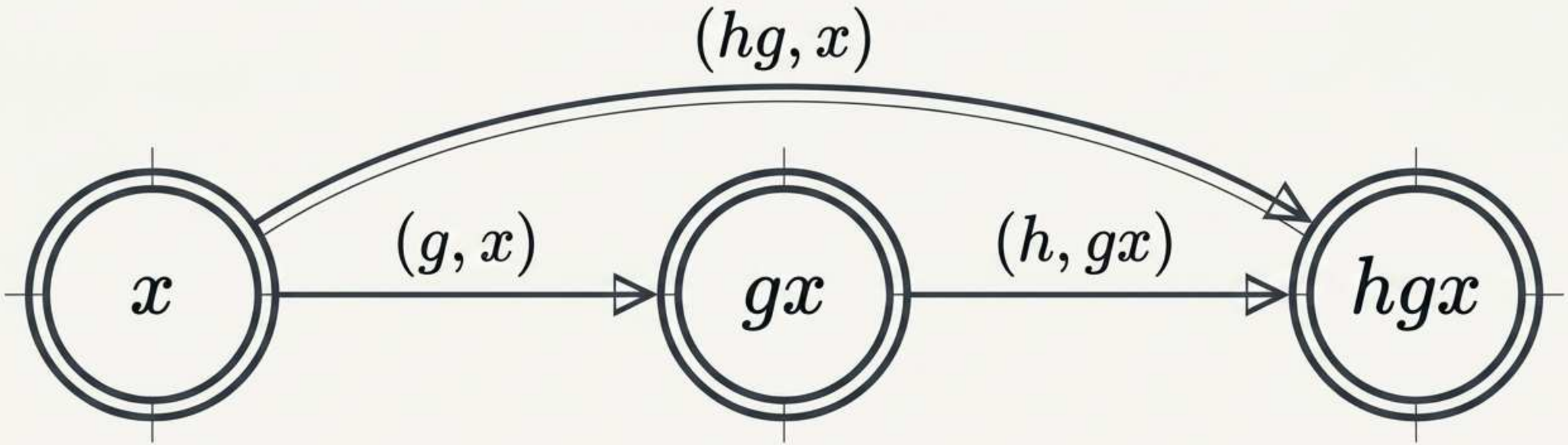
Higgs Mechanism

Local Fate: The same broken degrees of freedom are "eaten" by the local transport structure, altering the degrees of freedom of the gauge connection (massive vector bosons).

Conclusion: Symmetry breaking cannot be reduced to a purely algebraic replacement ($G \to H$). The exact same pair (G, H) has completely different physical consequences depending on whether the symmetry acts globally or requires local transport.

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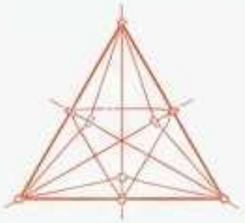
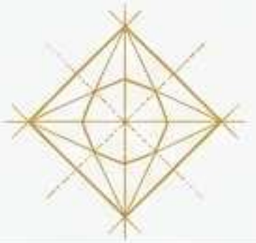
The Action Groupoid retains the source, target, and state data that bare groups suppress



- **Boundary-Restricted Composition**: (h, gx) composed with $(g, x) = (hg, x)$. Morphisms have distinct sources and targets, composing only when those boundaries match.
- **State-Relative Isotropy**: At the chosen vacuum v, transformations in H are endomorphisms ($v \to v$). Transformations outside H generate arrows to other orbit points ($v \to gv$).
- **Distinction**: Distinction is not absolute; it is induced relative to a constraint structure.

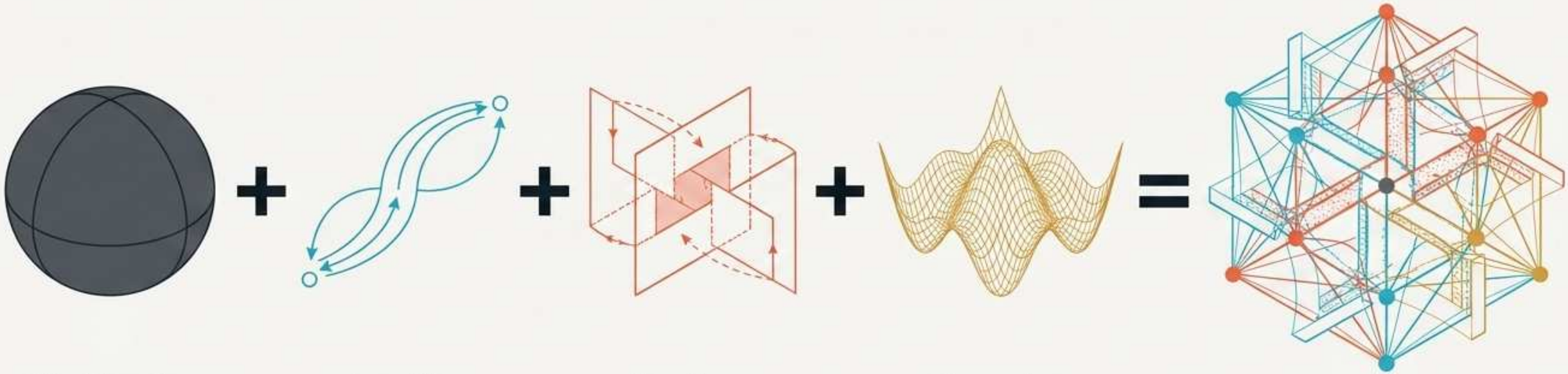
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The Compression Matrix: Knowing what data is retained and what is discarded

The Group (Compressed)		The Groupoid (Decompressed)	
	**Composition** Globally composable everywhere.		**Boundary-restricted** (composes only when target matches source).
	**Transport & Path** Suppresses the path taken.		**Drafting Cyan** Retains path history and transport rules.
	**Trivialization** Assumes a single global trivialization.		**Vermilion** Allows non-trivial gluing and structural obstructions.
	**Isotropy** Assumes uniform isotropy.		**Ochre** Retains state-relative isotropy and distinct stabilizers.

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Three distinct physical phenomena, one unified pattern of information retrieval



The Baseline:
Abstract Symmetry Group G

Locality:
Source/Target & Path Transport

Topology:
Transition Cocycles & Gluing Data

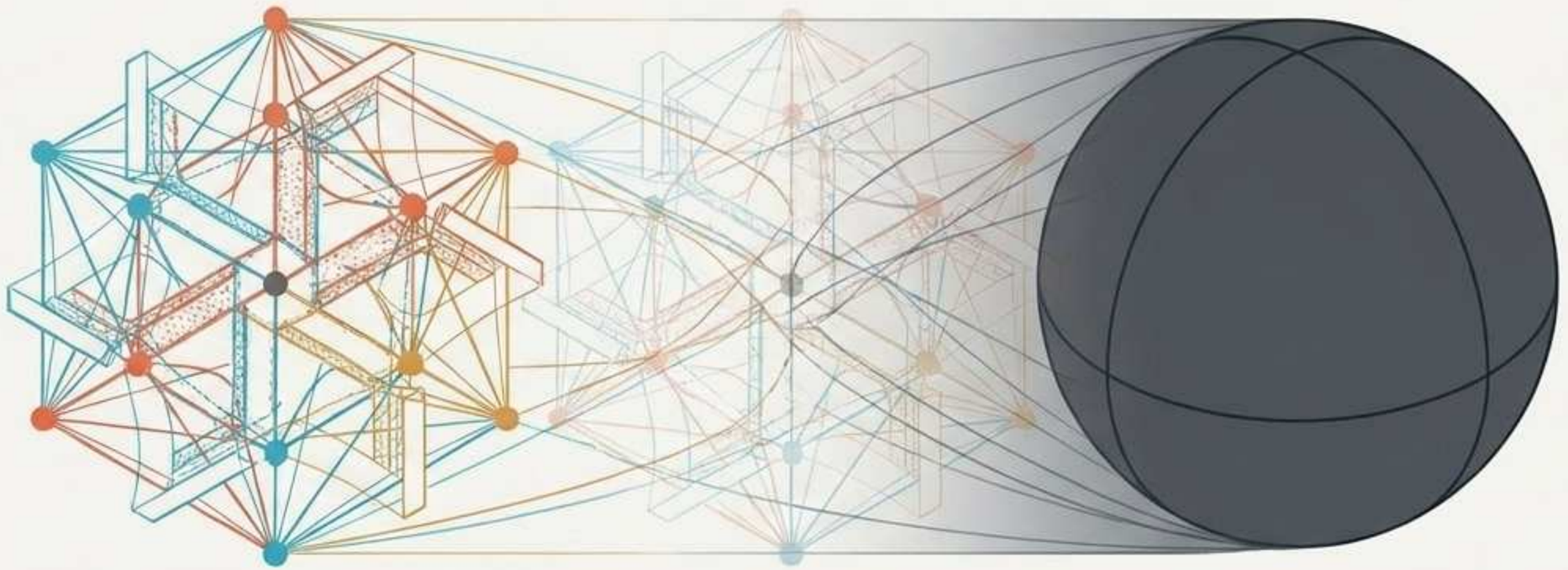
Dynamics:
Vacuum Selection & State-Relative Isotropy

The Admissibility Groupoid

Synthesis Statement: These are not three random mathematical facts. They are three independent reasons why abstract groups must be decompressed to reveal the specific physical architecture of where transformations act and how they compose.

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Which distinctions may be quotiented away without changing what the system can physically distinguish?



The Heuristic: When source, target, location, and trajectory are irrelevant, a group is exactly the right object.

The Reality: When those distinctions survive, the richer compositional structure of the groupoid is **not** additional decoration—it is the vital information the quotient would have erased.

Compression is not an error, provided you know exactly what data you are choosing to forget.

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